

TIAN SHAN, Kyrgyzstan

WMO: 369820

Lat: 41.897N

Lon: 78.185E

Elev: 11939

StdP: 9.37

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.9	-19.2	-32.5	1.4	-18.8	-28.4	1.8	-16.7	20.7	5.1	18.3	7.0	1.4	50	0.916

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	57.9	43.2	55.2	41.7	52.5	40.3	47.6	52.0	44.7	50.1	42.6	48.9	9.1	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
45.8	71.8	49.5	42.6	63.4	46.5	39.9	57.1	44.1	23.4	51.7	21.4	50.1	20.0	48.9	59.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.9	17.2	14.8	-27.6	64.8	3.6	3.8	-30.2	67.5	-32.3	69.8	-34.3	71.9	-36.8	74.7		
(5)			-28.2	49.0	3.3	4.8	-30.6	52.5	-32.6	55.3	-34.4	58.0	-36.9	61.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	21.9	-1.8	1.4	12.8	23.5	31.5	38.5	42.6	41.6	34.9	23.5	10.8	1.8		
	DBStd	16.92	7.15	7.78	8.19	7.57	5.46	3.94	4.11	4.04	5.29	6.65	7.98	6.59		
	HDD50	10268	1605	1361	1154	794	573	344	233	262	452	820	1176	1494		
	HDD65	15739	2070	1781	1619	1244	1038	794	695	726	902	1285	1626	1959		
	CDD50	4	0	0	0	0	0	0	3	1	0	0	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	4.9	3.4	3.7	4.5	5.3	5.6	5.8	5.8	5.4	5.9	5.4	4.0	3.7		
	PrecAvg	11.5	0.1	0.2	0.5	0.9	1.5	2.1	2.0	1.7	1.1	0.5	0.3	0.3		
(15) Precipitation	PrecMax	17.3	0.4	0.8	1.2	2.3	5.1	4.1	4.1	3.8	2.4	1.7	0.9	0.9		
	PrecMin	4.9	0.0	0.0	0.2	0.2	0.2	0.5	0.4	0.0	0.0	0.0	0.0	0.0		
	PrecStd	2.9	0.1	0.2	0.2	0.6	0.9	0.9	0.9	1.1	0.6	0.4	0.2	0.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.3	26.1	36.9	47.4	52.1	56.8	62.4	60.9	55.2	46.5	34.9	22.7		
		MCWB	15.0	20.8	28.3	34.9	38.0	41.2	47.1	43.7	41.7	32.1	25.9	17.9		
	2%	DB	15.6	21.2	32.8	42.6	47.8	52.7	58.8	57.5	51.7	41.2	30.3	19.1		
		MCWB	12.2	17.4	25.7	32.1	36.2	39.2	43.3	42.9	37.8	31.3	24.3	14.9		
	5%	DB	12.8	17.7	29.5	39.3	44.6	50.2	55.9	54.7	49.1	37.7	26.6	15.8		
		MCWB	10.1	14.5	23.9	30.7	34.3	38.6	42.3	41.7	36.6	29.3	21.6	12.5		
	10%	DB	10.1	14.4	26.6	35.8	41.4	47.6	53.0	52.0	46.2	34.6	23.1	13.1		
		MCWB	7.7	11.7	22.2	28.9	33.1	37.7	41.3	40.5	35.5	27.6	19.3	10.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.8	21.4	29.5	38.6	43.7	46.4	54.2	52.7	47.3	36.5	28.2	19.3		
		MCDB	19.2	24.4	34.3	42.1	47.1	49.6	58.6	56.4	51.5	38.7	31.5	21.8		
	2%	WB	12.6	17.7	26.8	34.4	38.0	43.0	48.2	47.3	42.7	32.3	24.9	15.2		
		MCDB	15.1	20.8	31.5	40.1	43.9	47.2	52.4	52.1	46.1	39.1	28.9	18.3		
	5%	WB	10.2	14.7	24.5	31.5	35.8	40.8	44.9	44.3	38.7	30.2	22.0	12.8		
		MCDB	12.6	17.5	29.0	37.2	41.7	46.7	50.9	50.1	44.5	36.4	26.0	15.6		
	10%	WB	7.7	11.8	22.1	29.5	34.0	39.0	42.6	42.1	36.2	28.4	19.3	10.4		
		MCDB	10.0	14.2	26.3	35.0	40.1	45.7	50.2	48.5	43.8	33.5	23.1	13.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	20.7	21.7	21.3	18.8	16.4	16.5	18.3	18.9	18.8	18.3	19.8	19.6		
		MCDBR	22.3	22.7	21.6	22.1	21.3	20.9	23.3	24.1	23.4	21.6	22.0	21.7		
		MCWBR	19.5	19.2	16.3	14.2	11.5	10.6	11.1	12.3	12.2	13.3	16.5	18.2		
	5% WB	MCDBR	21.8	22.2	21.3	19.7	19.2	18.4	20.2	20.7	22.2	19.5	20.8	20.8		
		MCWBR	19.3	19.2	17.0	14.3	12.7	12.4	13.5	13.8	15.6	13.8	16.6	18.0		
(40) Clear-Sky Solar Irradiance	taub		0.216	0.245	0.304	0.348	0.342	0.332	0.313	0.315	0.296	0.274	0.241	0.209		
	taud		2.377	2.293	2.133	2.061	2.173	2.286	2.384	2.374	2.379	2.356	2.390	2.424		
	Ebn at Noon		307	309	298	291	295	297	302	299	299	293	292	302		
	Edh at Noon		25	32	43	50	46	42	37	36	33	30	24	22		
(44) All-Sky Solar Radiation	RadAvg		669	924	1267	1563	1840	1932	1956	1779	1450	1032	706	575		
	RadStd		58	78	91	139	121	100	102	95	102	80	67	47		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (4 neighbors)	+0.77	N/A	N/A	N/A	-1.24	-1.53	-365	-454	+250	+130	

Nomenclature: See separate page